

National University of Computer and Emerging Sciences, Lahore Campus



Course:
Program:
Submission Date:
Section:
Assignment no. :
Total Marks:

Data Mining
BSE, BCS
04-05-26
6A, 8A
05
100

Important Instructions:

- Use A4 sheets or narrow line sheets for solving assignments.
- Assignments done on register pages will be cancelled.
- Must attach the assignment title page shared to you at the end of assignment.
- This assignment must be completed independently by the student. The use of AI tools (including ChatGPT, Bard, Copilot, or any other generative AI system) is strictly prohibited.
- Students are expected to show all manual calculations and intermediate steps clearly.

NOTE: IT IS VERY IMPORTANT TO DO IT WITH YOUR OWN HANDS TO PREPARE YOURSELF FOR FINAL EXAMS.

Customer	V1: Age	V2: Annual Income (k\$)	V3: Spending Score (1-100)
A	15	20	80
B	18	30	75
C	40	60	50
D	45	65	45
E	70	15	10
F	65	20	15
G	60	50	15
H	65	45	18
I	15	10	40
J	60	50	45

Question 1: K-Means, K-medoids, K-modes

Using the above dataset cluster the points where $K=3$ using:

- K-Means are A, C and J.
- K-Medoids are F, H, J.

iii. K-Mode are A, B, C (if required)

Also explain which one theoretically and practically is better than the other two and why?

Question 3: Hierarchical Clustering

The table given is a distance matrix for 6 objects

	A	B	C	D	E	F
A	0					
B	0.12	0				
C	0.51	0.25	0			
D	0.84	0.16	0.14	0		
E	0.28	0.77	0.70	0.45	0	
F	0.34	0.61	0.93	0.20	0.67	0

1. Show the result - step by step (clusters, matrix, dendrogram), of hierarchical clustering using:

- I. Single link
- II. Complete link
- III. Average link

At the end do draw the final clustering figure and dendrogram.

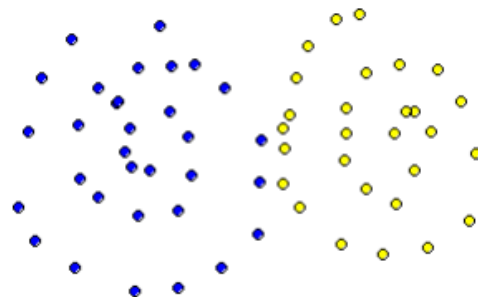
Question 4: Which clustering method will be used ?

Which clustering method(s) is most likely to produce the following results at $k = 2$? Choose the most likely method(s) and briefly explain why it/they will work better where others will not in at most 3 sentences. Here are the five clustering methods you can choose from: -

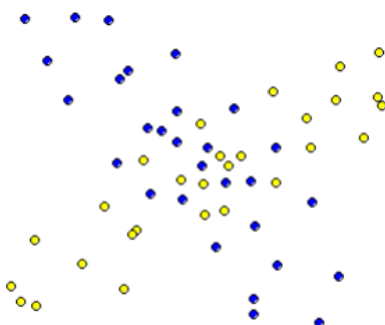
- Hierarchical clustering with single link
- Hierarchical clustering with complete link
- Hierarchical clustering with average link
- K-means
- K-medoids
- K-Modes



b)



c)



Question 5: Birch Algorithm

Make a dataset of minimum 2 attributes and 20 rows (values must not be concentrated around one point) and cluster the points using birch algorithm, by taking a reasonable parameter of cluster such that atleast touch 3 levels of tree.

Question 6: CLARA & CLARANS

A large e-commerce company is analyzing a dataset of approximately 500,000 customers to segment them into meaningful groups based on their purchasing behavior. Each customer is represented by a feature vector $X = (\text{Annual Spending}, \text{Visit Frequency}, \text{Time Spent on App})$. The dataset is large in size and contains noise as well as outliers. Due to computational constraints and the need for efficient clustering, the data science team is considering two algorithms: CLARA Algorithm and CLARANS Algorithm. CLARA applies clustering on multiple samples of the dataset, while CLARANS performs a randomized search in the solution space to find optimal medoids.

Based on this scenario, explain how both algorithms would process such a large dataset and compare their performance in terms of computational efficiency, memory usage, and quality of clusters. Furthermore, justify why CLARA may produce faster results but potentially miss important patterns, whereas CLARANS may take more time but yield better clustering results. Finally, recommend which algorithm should be used if the company has limited computational resources and strict time constraints, and which one should be preferred if achieving higher clustering accuracy is the primary objective.

Question 7: Reasoning

1. Why do K-Means fail on **non-spherical clusters**?
2. What are the limitations of hierarchical clustering?
3. Explain the scenario where we have to use k-medoid and where we have to use CLARA?
4. Why is CLARA needed?

National University of Computer and Emerging Sciences
Department of Computer Science, School of Computing

DATA MINING



Assignment # _____

Section: _____

Submitted by:

Name: _____

Roll no: _____

Submitted to: Dr. Muhammad Aasim Qureshi

Date Assigned: _____

Date Submission: _____